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Electromagnetic Induction

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	32 questions
Answer key	at the end

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MCQ (Single Correct Answer) 32 questions

Q1. RE-NEET 2026

Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is :

Open solution

(A) $2\mu_0\pi n^2r^2l$

(B) $\mu_0\pi n^2r^2l$

(C) $\mu_0 n^2r^2l$

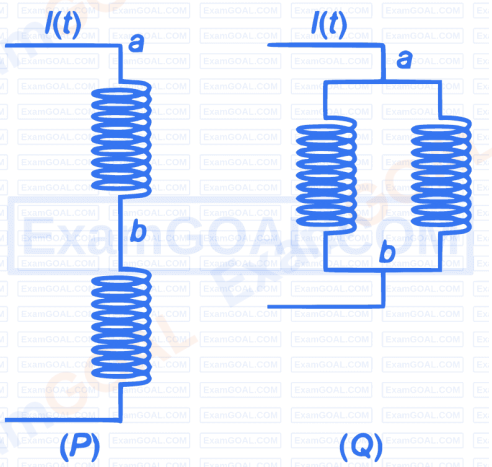
(D) $(\mu_0/2\pi)n^2r^2l$

Q2. RE-NEET 2026

Two identical inductors are connected in two different configurations P and Q, where a time varying current $l(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is:

Open solution

[Neglect the effect of mutual inductance.]



(A) 2

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 1

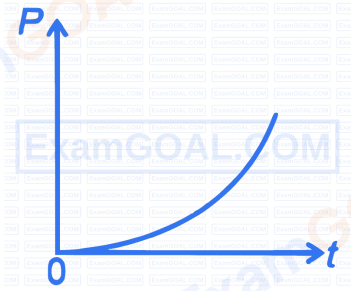
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Q3. RE-NEET 2026

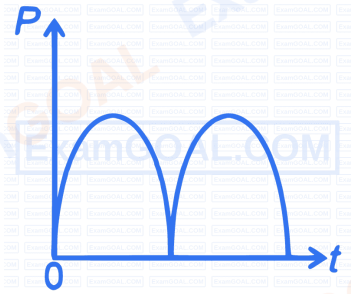
Open solution

A conducting loop of finite resistance lies on the $x - y$ plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is:

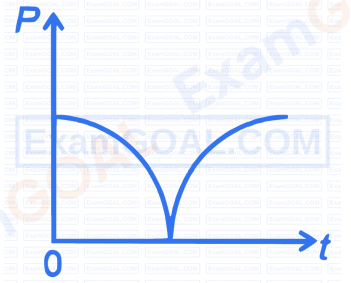
(A)



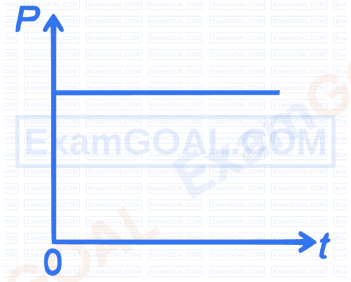
(B)



(C)



(D)



Q4. NEET 2026

Open solution

A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is $2\ \text{cm s}^{-1}$, in a direction normal to the shorter side of the loop, will be :

(A) 4.8×10^{-4} volt (B) 1.2×10^{-4} volt (C) 1.3×10^{-4} volt (D) 1.8×10^{-4} volt

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Q5. NEET 2025

AB is a part of an electrical circuit (see figure). The potential difference " $V_A - V_B$ ", at the instant when current $i = 2\ \text{A}$ and is increasing at a rate of $1\text{amp}/$ second is:



Open solution

(A) 9 volt (B) 10 volt (C) 5 volt (D) 6 volt

Q6. NEET 2024 (Re-Examination)

Let us consider two solenoids A and B, made from same magnetic material of relative permeability μ_r and equal area of cross-section. Length of A is twice that of B and the number of turns per unit length in A is half that of B. The ratio of self inductances of the two solenoids, $L_A : L_B$ is

Open solution

(A) $1:2$ (B) $2:1$ (C) $8:1$ (D) $1:8$

Q7. NEET 2023 Manipur

An emf is generated by an ac generator having 100 turn coil, of loop area $1\ \text{m}^2$. The coil rotates at a speed of one revolution per second and placed in a uniform magnetic field of $0.05\ \text{T}$ perpendicular to the axis of rotation of the coil. The maximum value of emf is :-

Open solution

(A) 3.14 V (B) 31.4 V (C) 62.8 V (D) 6.28 V

Q8. NEET 2023

The net magnetic flux through any closed surface is :

Open solution

(A) Positive (B) Infinity (C) Negative (D) Zero

Q9. NEET 2022 Phase 2

The magnetic flux linked to a circular coil of radius R is

$\phi = 2t^3 + 4t^2 + 2t + 5$ Wb

The magnitude of induced emf in the coil at t = 5 s is

Open solution

(A) 192 V (B) 108 V (C) 197 V (D) 150 V

Q10. NEET 2022 Phase 1

A square loop of side 1 m and resistance 1 Ω is placed in a magnetic field of 0.5 T. If the plane of loop is perpendicular to the direction of magnetic field, the magnetic flux through the loop is

Open solution

(A) 2 weber (B) 0.5 weber (C) 1 weber (D) Zero weber

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Q11. NEET 2021

Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 >> R_2$, the mutual inductance M between them will be directly proportional to :

(A) $\frac{R_2^2}{R_1}$ (B) $\frac{R_1}{R_2}$ (C) $\frac{R_2}{R_1}$ (D) $\frac{R_1^2}{R_2}$

Open solution

Q12. NEET 2019

A 800 turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field 5×10^{-5} T. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s, the emf induced in the coil will be :

(A) 0.2 V (B) 0.02 V (C) 2 V (D) 2×10^{-3} V

Open solution

Q13. NEET 2019

In which of the following devices, the eddy current effect is **not** used?

(A) magnetic braking in train (B) electromagnet

(C) electric heater (D) induction furnace

Open solution

Q14. NEET 2017

A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is 10 π^2 Ω, the total charge flowing through the coil during this time is

(A) 16 μC (B) 32 μC (C) 16π μC (D) 32π μC

Open solution

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Q15. NEET 2016 Phase 2

A uniform magnetic field is restricted within a region of radius r. The magnetic field changes with time at a rate $\frac{d\vec{B}}{dt}$. Loop 1 of radius R > r encloses the region r and loop 2 of radius R is outside the region of magnetic field as shown in the figure. Then the e.m.f. generated is

Open solution



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(A) zero in loop 1 and zero in loop 2

(B) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 2

(C) $-\frac{d\vec{B}}{dt}\pi R^2$ in loop 1 and zero in loop 2

(D) $-\frac{d\vec{B}}{dt}\pi r^2$ in loop 1 and zero in loop 2

Q16. NEET 2016 Phase 1

A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is

Open solution

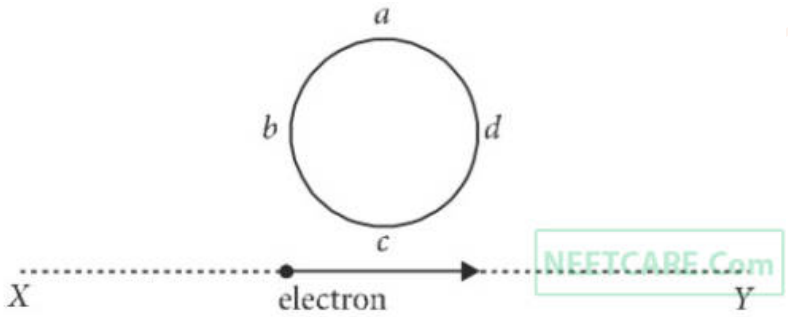
(A) 2 H (B) 1 H (C) 4 H (D) 3 H

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Q17. AIPMT 2015

An electron moves on a straight line path XY as shown. The abcd is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil ?

Open solution



(A) The current will reverse its direction as the electron goes past the coil

(B) No current induced

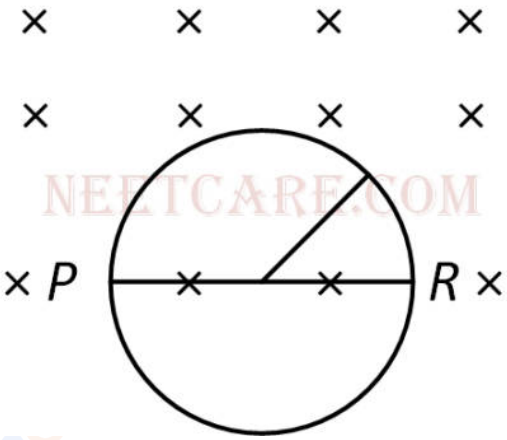
(C) abcd

(D) adcb

Q18. AIPMT 2014

A thin semicircular conducting ring (PQR) of radius r is falling with its plane vertical in a horizontal magnetic field B, as shown in the figure.

Open solution



The potential difference developed across the ring when its speed is v, is

(A) zero

(B) $\frac{Bv\pi r^2}{2}$ and P is at higher potential

(C) πrBV and R is at higher potential

(D) 2rBV and R is at higher potential

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Q19. NEET 2013 (Karnataka)

A current of 2.5 A flows through a coil of inductance 5 H. The magnetic flux linked with the coil is

(A) 0.5 Wb (B) 12.5 Wb (C) zero (D) 2 Wb

Open solution

Q20. NEET 2013

A wire loop is rotated in a magnetic field. The frequency of change of direction of the induced e.m.f. is

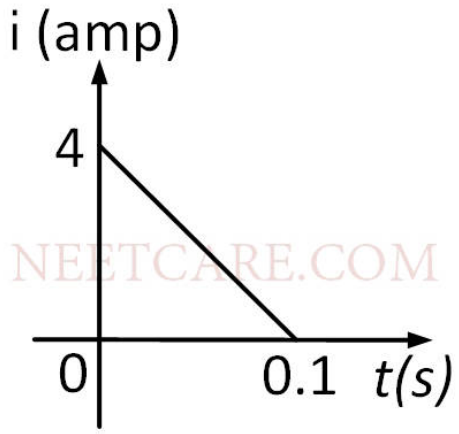
(A) four times per revolution (B) six times per revolution

(C) once per revolution (D) twice per revolution

Open solution

Q21. AIPMT 2012 Mains

In a coil of resistance 10 Ω, the induced current developed by changing magnetic flux through it, is shown in figure as a function of time. The magnitude of change in flux through the coil in weber is



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(A) 8 (B) 2 (C) 6 (D) 4

Open solution

Q22. AIPMT 2012 Prelims

A coil of resistance 400 Ω is placed in a magnetic field. If the magnetic flux ϕ (Wb) linked with the coil varies with time t (sec) as $\phi = 50t^2 + 4$.

The current in the coil at t = 2 sec is

(A) 0.5 A (B) 0.1 A (C) 2 A (D) 1 A

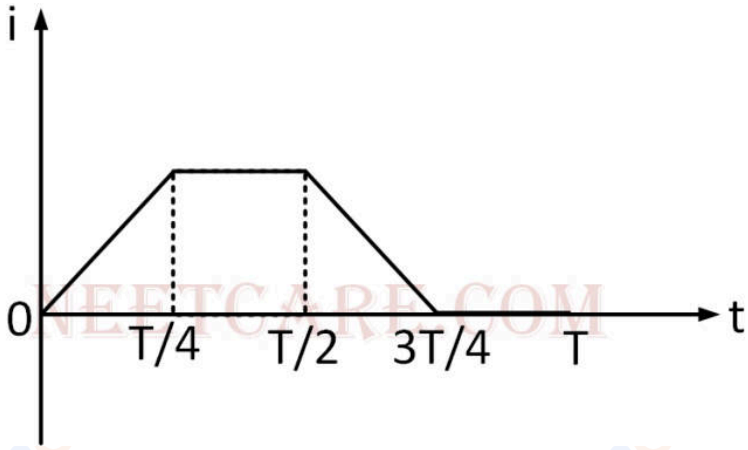
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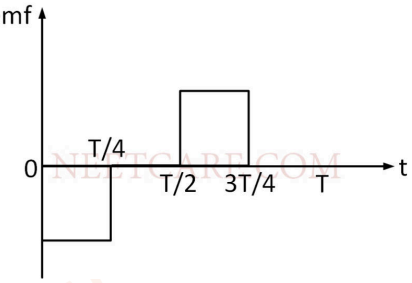
Q23. AIPMT 2011 Prelims

The current i in a coil varies with time as shown in the figure. The variation of induced emf with time would be

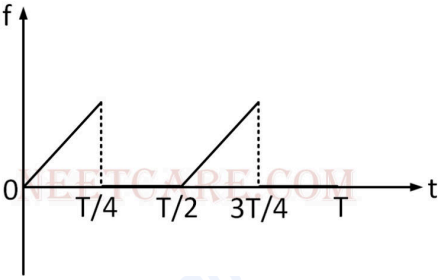
Open solution



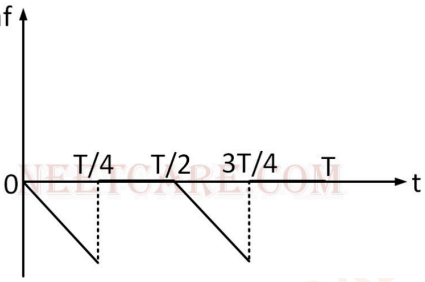
(A)



(B)

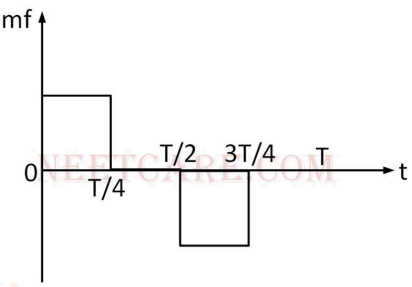


(C)



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(D)



Q24. AIPMT 2010 Prelims

A conducting circular loop is placed in a uniform magnetic field, B = 0.025 T with its plane perpendicular to the loop. The radius of the loop is made to shrink at a constant rate of 1 mm s^{-1}. The induced emf when the radius is 2 cm, is

Open solution

(A) $2\pi\mu V$ (B) $\pi\mu V$ (C) $\frac{\pi}{2}\mu V$ (D) $2\mu V$

Q25. AIPMT 2009

A rectangular, a square, a circular and an elliptical loop, all in the (x-y) plane, are moving out of a uniform magnetic field with a constant velocity. $\vec{V} = v\hat{i}$. The magnetic field is directed along the negative z axis direction. The induced emf, during the passes of these loops, out of the field region, will not remain constant for

Open solution

(A) The circular and the elliptical loops

(B) only the elllliptical loop

(C) any of the four loops

(D) the rectangular, circular and elliptical loops

Q26. AIPMT 2009

A conducting circular loop is placed in a uniform magnetic field 0.04 T with its plane perpendicular to the magnetic field. The radius of the loop starts shrinking at 2 mm/s. The induced emf in the loop when the radius is 2 cm is

Open solution

(A) 4.8$\pi\mu$V (B) 0.8$\pi\mu$V (C) 1.6$\pi\mu$V (D) 3.2$\pi\mu$V

Q27. AIPMT 2008

A long solenoid has 500 turns. When a current of 2 ampere is passed through it, the resulting magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is

Open solution

(A) 1.0 henry (B) 4.0 henry (C) 2.5 henry (D) 2.0 henry

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Q28. AIPMT 2008

A circular disc of radius 0.2 meter is placed in a uniform magnetic field of induction $\frac{1}{\pi}\left(\frac{Wb}{m^2}\right)$ in such a way that its axis makes an angle of 60° with $\vec{B}$. The magnetic flux linked with the disc is

(A) 0.08 Wb (B) 0.01 Wb (C) 0.02 Wb (D) 0.06 Wb

Open solution

Q29. AIPMT 2006

Two coils of self inductance 2 mH and 8 mH are placed so close together that the effective flux in one coil is completely linked with the other. The mutual inductance between these coils is

(A) 16 mH (B) 10 mH (C) 6 mH (D) 4 mH

Open solution

Q30. AIPMT 2005

As a result of change in the magnetic flux linked to the closed loop as shown in the figure, an e.m.f. V volt is induced in the loop. The work done (joule) in taking a charge Q coulomb once along the loop is



(A) QV (B) 2QV (C) QV/2 (D) zero

Open solution

Q31. AIPMT 2004

The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in a time Δt. Then the total quantity of electric charge Q that passes any point in the circuit during the time Δt is represented by

(A) $Q = \frac{1}{R}.\frac{\Delta\phi}{\Delta t}$ (B) $Q = \frac{\Delta\phi}{R}$

(C) $Q = \frac{\Delta\phi}{\Delta t}$ (D) $Q = R.\frac{\Delta\phi}{\Delta t}$

Open solution

Q32. AIPMT 2001

For a coil having L = 2 mH, current flow through it is $I = t^2e^{-t}$ then, the time at which emf become zero

(A) 2 sec (B) 1 sec (C) 4 sec (D) 3 sec

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Answer Key

1. **B**	2. **A**	3. **C**	4. **D**	5. **B**
6. **A**	7. **B**	8. **D**	9. **A**	10. **B**
11. **A**	12. **B**	13. **C**	14. **B**	15. **D**
16. **B**	17. **A**	18. **D**	19. **B**	20. **D**
21. **B**	22. **A**	23. **A**	24. **B**	25. **A**
26. **D**	27. **A**	28. **C**	29. **D**	30. **A**
31. **B**	32. **A**			

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